THE PAPER PLANE CHALLENGE

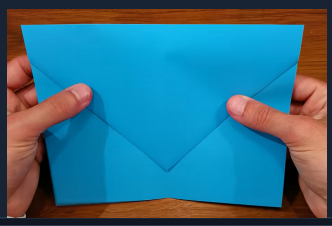
Three-page engineering report • Objective: maximise airborne duration, not simply distance.

Design choice. Use a broad-wing, light paper glider with a stiff folded nose, slight dihedral and tiny adjustable trailing-edge tabs. Broad wings aim to reduce wing loading and support a slow, shallow glide. This is a proposed design, not a physically tested result.

Starting dimensions

Feature	Target	Reason
Sheet	A4: 210 × 297 mm	One standard sheet
Fuselage	≈ 297 mm centre fold	Straight reference and stiffness
Wings	Near-full sheet span; 65–85 mm broad chord	More lifting area at low speed
Nose	3–5 tight paper folds	Stiffens nose and moves CG forward
Trim tabs	≈ 5 mm upward, symmetric	Fine pitch adjustment

Construction sequence



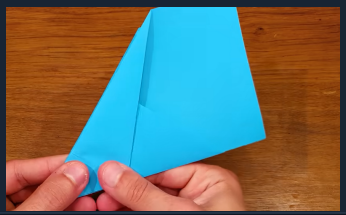
Step 1 — Fold the top corners to the centre

Hold the A4 sheet in front of you and fold both top corners in so the creases meet in a V at the centre line. Crease firmly. (Optional: fold the sheet lengthwise in half first and reopen to mark the centre line.)



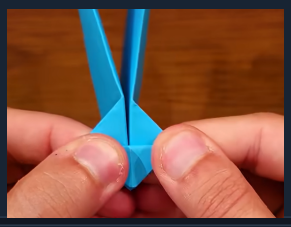
Step 2 — Fold up the nose tip

Fold the point of the nose up over itself to form a small triangular tip and crease it hard. This locks the layers together and adds weight forward for a stiff nose and forward CG.



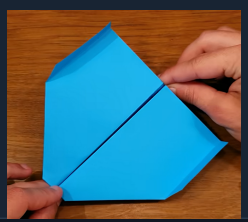
Step 3 — Fold in half along the centre crease

Fold the plane in half along the centre crease with the nose folds on the outside. Line up the edges and the nose so both sides are symmetrical.



Step 4 — Pinch and tighten the nose

Pinch the nose fold tightly between thumb and forefinger so the layers sit together. Aim for 3–5 tight paper folds at the nose; the wings stand upright while you work.



Step 5 — Fold the wings down and open them out

Fold each wing down symmetrically, keeping the wings broad (65–85 mm chord). Open them flat with the tips slightly higher than the centre (dihedral). Fold the wing tips up about 5 mm as matching trailing-edge tabs. Check symmetry, then launch gently and nearly level.

AERODYNAMICS AND DESIGN REASONING

How lift is generated

The wing deflects air downward, producing lift (perpendicular to the airflow) and drag (opposing motion). Lift is modelled as:

$$L = \tfrac{1}{2} \rho V^2 S C_L$$

Here ρ is air density, V is airspeed, S is wing area and C_L is the lift coefficient.

Forces during launch, glide and landing

Launch: the hand gives a short impulse; there is no thrust. Release smoothly and nearly level to avoid a stall.

Glide: lift supports the weight while drag removes energy, so the plane descends as it moves forward.

Landing: speed and lift fall; a steep dive suggests a forward CG or poor trim.

Wing area, wing loading and stability

Larger wing area lowers wing loading (W/S) and allows slower flight. The folded nose moves the CG forward for pitch stability, and slight dihedral with equal trim tabs keeps the glide steady.

Why a glider may stay up longer than a dart

A dart is tuned for speed and distance. A broad-wing glider targets a low sink rate; **minimum sink** is the main goal for longest time aloft, not maximum glide distance.

Assumptions and limitations

Single A4 sheet, hand launch, no added materials, still indoor air. Design dimensions are starting points and need testing to validate performance.

References

1. NASA Glenn Research Center, lift and aerodynamics. 2. Anderson, J. D., *Fundamentals of Aerodynamics*, McGraw-Hill. 3. NASA, four forces of flight. 4. “Problem 2 — The Paper Plane Challenge” brief.

EXPERIMENT, RESULTS AND BONUS

Baseline method

Test indoors with still air. Use the same launcher, release height, launch angle, paper type and timing method. Time from release to first floor contact. Keep the final design unchanged across five trials. Note stalls, dives, turns or invalid releases.

Five-trial flight log — enter actual measurements

Measured mean = 2.8 s | range 2.4–3.3 s

Trial	Duration (s)	Notes
1	2.4	
2	2.8	
3	3.3	
4	2.7	
5	2.8	

Modifications and fair test

A — elevator trim: compare neutral tabs with both raised by 2 mm; use five trials per setting and change no other variable. **B — dihedral:** increase wing-tip rise by 3–5° per side and repeat five trials, holding launch, CG and wing area as constant as possible. Compare means and ranges. Optionally test one extra nose fold, but note that this changes both CG and nose stiffness.

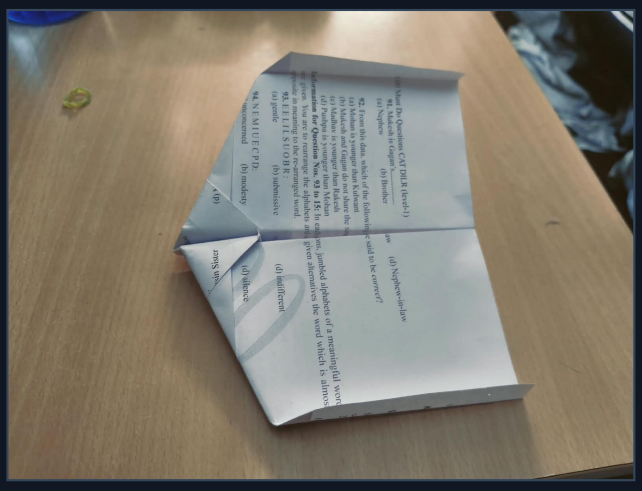
For a fair test, use the same operator, room, release point, paper and timing method. Alternate conditions where possible. Repeat only clearly invalid trials and document why.

Bonus: compare a second design

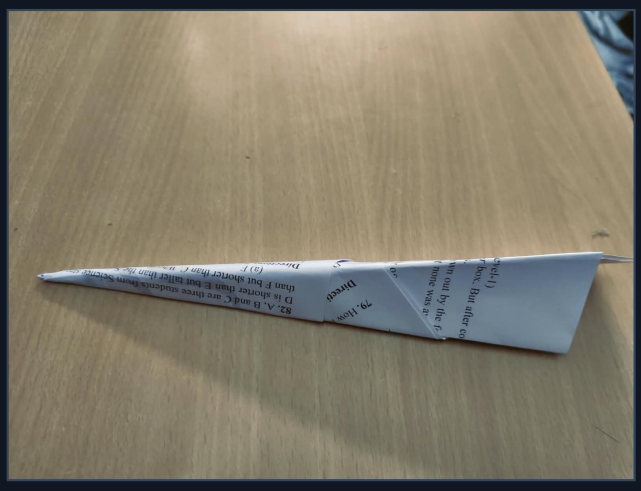
Build a conventional narrow dart from a second A4 sheet. Test both designs with the same conditions and five trials each. Do not conclude which is better until the measured means are compared.

Design	T1	T2	T3	T4	T5	Mean
Broad-wing glider	2.4	2.8	3.3	2.7	2.8	2.8
Conventional dart	1.9	2.0	2.1	1.9	1.8	1.94

Evidence: finished designs



Broad-wing glider: broad wings, nose folds and trailing-edge tabs.



Conventional dart: narrow, folded design (side view).